(a) Candidate Pool Budget K & Latency Trade-off

End-to-End HR@10 (%)

Inference Latency (ms)

Candidate Pool Budget (K)

Pareto Optimal ($K = 50$)

CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Latency (ms/sample)

(b) RRF Smoothing Parameter μ across City Densities

End-to-End HR@10 (%)

$\mu = 15$ **(NYC/TKY Opt)**

$\mu = 50$ **(CA Opt)**

RRF Smoothing Constant (μ)

CA (Sparse, Large-scale)
NYC (Compact, Dense)
TKY (High-density Network)

(c) Semantic Cluster Number K_c vs. Long-tail Generalization

Long-tail POI HR@10 (%)

Non-singleton Bucket Coverage (%)

Optimal $K_c = 256$

Semantic Cluster Vocabulary Capacity (K_c)

CA Long-tail HR@10
NYC Long-tail HR@10
TKY Long-tail HR@10
Bucket Coverage (%)

(d) Profile Token Length L_{prof} vs. Output Robustness

End-to-End HR@10 (%)

Valid JSON Parse Rate (%)

Optimal $L_{prof} = 120$

Max Profile Token Length (L_{prof})

CA End-to-End HR@10
NYC End-to-End HR@10
TKY End-to-End HR@10
Valid JSON Parse Rate (%)